SOCIAL NETWORKS

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Week 11 — Question Paper (Comprehension-Based MCQ / MSQ)

Comprehension 1 — Six Hops to Anyone: The Small-World Effect

Case Study

A researcher, Leila, revisits a classic question on a modern messaging app: if you pick two strangers anywhere in the world, how many people stand between them in the chain of acquaintances? To test it she seeds many participants with a token addressed to a distant target person and asks each holder to forward it only to a single acquaintance they believe is somehow closer to the target — never directly to the target. She then records, for the tokens that arrive, how many forwarding steps the chain took.

Most arriving chains turn out to be remarkably short — only a handful of steps on average. At first this seems impossible: the world holds billions of people, so surely a stranger must be very far away. Yet Leila reasons about reach level by level: if each person has, say, a hundred acquaintances, and each of those has a hundred more, the tally he can reach swells many times over with each new level, so the whole population is covered after only a few of them. The world is enormous and tiny at the same time.

To explain why such short chains exist, Leila draws on two ideas. People mostly befriend others who are near them or like them, which packs the network into tight local clusters. But a few of everyone's ties reach far outside their immediate circle — to a different city, profession, or country. It is this small minority of long-range ties, laid over the dense local clustering, that stitches the distant clusters together and makes the whole world reachable in only a few hops.

Questions 1–10

Q1. [MCQ] What did the original version of this experiment (Milgram's) find about the chains that reached the target?

- **A.** They reached the target through a direct connection in fewer than three steps every time.
- **B.** They showed there was no measurable typical number of steps between two people.
- **C.** They took about six forwarding steps on average, the “six degrees” result.
- **D.** They reached the target only after passing through more than a hundred steps.

Answer: C

Reference: Lecture 144 – Milgram's Experiments (≈00:08–end)

Solution: Milgram observed that the chains that arrived took roughly six steps on average — the celebrated “six degrees of separation.” They did not arrive in under three steps, the average was clearly measurable, and the count was around six rather than in the hundreds.

Q2. [MCQ] Why is the “only a few hops” result less surprising once Leila reasons level by level?

- **A.** Because the number of people reachable multiplies at every level, covering the population fast.
- **B.** Because a central directory quietly connects every participant straight to the target person.

C. Because almost everyone in the world already knows the target person personally to begin with.

D. Because each person has only one acquaintance, so the chain is forced to stay extremely short.

Answer: A

Reference: Lecture 144 – Milgram's Experiments (≈07:16–end)

Solution: Reach grows multiplicatively: a hundred acquaintances, each with a hundred more, gives ten thousand at two steps and so on, so the whole population is covered within a few levels. There is no central directory, strangers are not already known, and people have many acquaintances, not one.

Q3. [Match the following] Match each term in Column A with its description in Column B.

Column A (term)	Column B (description)
1. Small-world property	a. a long-range tie reaching outside one's local cluster
2. Homophily	b. forwarding using only knowledge of one's own contacts
3. Weak tie	c. any two nodes are joined by a surprisingly short path
4. Decentralized search	d. the tendency to befriend nearby or similar people

A. 1-c, 2-d, 3-a, 4-b

B. 1-d, 2-c, 3-b, 4-a

C. 1-c, 2-a, 3-d, 4-b

D. 1-c, 2-d, 3-b, 4-a

Answer: A (1-c, 2-d, 3-a, 4-b)

Reference: Lectures 143–147 – Small-world effect, homophily, weak ties, search

Solution: The small-world property is the short-path fact (1-c); homophily is befriending nearby or similar people (2-d); a weak tie is a long-range link out of one's cluster (3-a); and decentralized search forwards using only local knowledge (4-b).

Q4. [MCQ] According to Watts and Strogatz, which two ingredients together produce the small-world effect?

A. Dense local clustering from homophily, plus a few long-range weak ties between clusters.

B. A complete graph in which every person is directly connected to every other person.

C. Purely random connections everywhere, with no local clustering of any kind at all.

D. A central hub that every single person in the world must connect to directly.

Answer: A

Reference: Lecture 145 – The Reason (≈00:07–end)

Solution: Watts and Strogatz attributed the effect to homophily, which creates tight local clusters, combined with a few weak ties that bridge those clusters. It does not require a complete graph, it is not purely random (clustering remains), and there is no universal central hub.

Q5. [MCQ] Leila runs the snippet below on her reconstructed acquaintance network. What does it report?

```
import networkx as nx
L = nx.average_shortest_path_length(G)
print(L)
```

A. The largest number of acquaintances that any single person in the network has.
B. The total count of people in the network, regardless of how they are connected.
C. The average number of hops on the shortest path between pairs of people.
D. The fraction of a person's acquaintances who are themselves acquainted.

Answer: C

Reference: Lecture 143 – Small World Effect: An Introduction (≈10:00–end)

Solution: average_shortest_path_length returns the mean shortest-path distance over all pairs of nodes — the typical number of hops, which the small-world effect predicts is small. It is not the maximum degree, not the node count, and not the clustering coefficient.

Q6. [Statement type] Consider the two statements about the small-world experiment and decide their truth.

Statement I: The experiment found an average of about six hops between two random people.
Statement II: This short average shows that the world's acquaintance network is disconnected.

A. Both statements are true.
B. Both statements are false.
C. Statement I is true, but Statement II is false.
D. Statement I is false, but Statement II is true.

Answer: C

Reference: Lectures 143–144 – Small-world effect and Milgram

Solution: The first statement is correct: the average chain length was about six. The second is wrong: a short average path actually shows the network is connected — reachable in few hops — not disconnected. So the first holds and the second fails.

Q7. [MCQ] What role do the few long-range weak ties play in producing short chains?

A. They lock everyone inside their own local cluster, lengthening the chains between clusters.
B. They bridge distant clusters, providing shortcuts that sharply reduce path lengths.
C. They duplicate the existing local ties, adding no new shortcuts to the network at all.
D. They disconnect the network into isolated pieces with no path between most people.

Answer: B

Reference: Lecture 145 – The Reason (≈04:55–end)

Solution: Weak ties act as bridges between otherwise distant clusters, creating shortcuts that collapse long local detours into a few hops. They do not trap people locally, they are not redundant copies of local ties, and they connect rather than disconnect the network.

Q8. [MCQ] What does homophily contribute to the structure of the acquaintance network?

A. A guarantee that no two acquaintances of a person are ever acquainted with each other.
B. A perfectly uniform network in which every person has an identical set of acquaintances.
C. Ties formed only with the most distant strangers, leaving no local connections at all.
D. Dense clusters of ties among nearby or similar people, rather than ties spread at random.

Answer: D

Reference: Lecture 145 – The Reason (≈02:47–end)

Solution: Homophily — befriending nearby or similar people — packs ties into dense local clusters rather than scattering them at random. It does not make everyone identical, it is not about distant strangers, and it tends to make a person's acquaintances acquainted, not the reverse.

Q9. [MSQ] Which statements about the small-world effect are correct? (Select all that apply.)

I. Two random people are typically separated by only a few hops.
II. Homophily creates dense local clustering in the network.
III. A few long-range weak ties act as shortcuts between clusters.
IV. The short chains prove the friendship network is disconnected.

A. I, II, IV
B. II, III, IV
C. I, III
D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 143–145 – Small-world effect and its reasons

Solution: Short separation (I), homophily-driven clustering (II), and weak-tie shortcuts (III) are all correct. Short chains show the network is connected, not disconnected, so IV is false.

Q10. [MCQ] Leila calls the result "counterintuitive but true." What is the counterintuitive contrast?

A. The world has only a few people, yet the chains between them are surprisingly long.
B. The world has billions of people, yet any two are separated by only about six hops.
C. The world is small in population, and accordingly the chains are also very short.
D. The world is large in population, and the chains are correspondingly enormous.

Answer: B

Reference: Lecture 144 – Milgram's Experiments (≈05:35–end)

Solution: The surprise is that despite billions of people — a huge world — any two are linked by only about six hops, so the world is at once big and small. The other options misstate either the population size or the chain length.

Comprehension 2 — Simulating a Small World and Routing Through It

Case Study

A developer builds the Watts–Strogatz generative model and then routes messages across it. The construction begins with a ring lattice: the nodes are placed on a cycle and each one is wired to a handful of its closest neighbours on the ring — the homophily links. A small fraction of edges is then rewired at random (equivalently, a few extra long-range edges are added), and these shortcuts are the weak ties that turn the lattice into a small-world network with short average path lengths.

On a richer two-dimensional grid, the long-range links are not placed uniformly. A node is joined to a far-away node with a probability that falls off with the grid distance d between them, governed by a clustering exponent: the chance of a link is proportional to 1/d raised to that exponent. Tuning the exponent controls how reachable distant nodes are, and the developer notes there is a particular value of the exponent at which routing on the grid works best.

Routing itself uses decentralized ("myopic") search. A node holding the message looks only at its own neighbours, estimates each neighbour's distance to the target along the lattice — without knowing its neighbours' long-range links — and forwards to whichever neighbour is closest to the target, repeating until the message arrives. Because it sees only local structure, this greedy routing need not trace the globally shortest path, yet across growing networks the number of hops it uses grows slowly. Key fragments:

```
import networkx as nx, random

# ring lattice of n nodes, each joined to its nearest neighbours, then rewired
G = nx.watts_strogatz_graph(n, k_near, p_rewire)

# myopic step: forward to the neighbour closest to the target on the lattice H
def best_neighbor(G, H, current, target):
    return min(G.neighbors(current),
               key=lambda x: nx.shortest_path_length(H, x, target))
```

Questions 1–10

Q1. [MCQ] How does the Watts–Strogatz construction begin, before any rewiring?

- **A.** With a complete graph in which every node is already joined to every other node.
- **B.** With a ring lattice in which each node is joined to a few of its nearest neighbours.
- **C.** With an empty graph of isolated nodes and no edges of any kind between them.
- **D.** With a star in which every node is joined only to one central hub node.

Answer: **B**

Reference: Lecture 146 – The Generative Model (≈01:11–end)

Solution: The model starts from a ring lattice — nodes on a cycle, each linked to a few nearest neighbours — representing local homophily ties. It does not start complete, empty, or as a star; the shortcuts are added only later by rewiring.

Q2. [MCQ] What does the random rewiring (a small rewiring probability) accomplish?

A. It removes all local ties, leaving a purely random graph with no clustering left.
B. It adds a few long-range shortcuts that sharply shorten the average path length.
C. It joins every pair of nodes, turning the lattice into a complete graph at once.
D. It leaves the lattice unchanged, since rewiring a few edges has no measurable effect.

Answer: B

Reference: Lecture 146 – The Generative Model (≈01:11–end)

Solution: Rewiring a small fraction of edges introduces a few long-range shortcuts (weak ties) that collapse long detours, giving the small-world property of short average paths while keeping most local clustering intact. It neither erases all local ties, nor completes the graph, nor leaves paths unchanged.

Q3. [MCQ] In the grid model, a node links to a far-away node with probability proportional to which quantity (d is the grid distance, q the clustering exponent)?

```
# probability that node u links to a far node v, distance d apart:
p_link  ∝  1 / (d ** q)
```

A. d raised to the power q, so farther nodes are more likely to be linked.
B. q raised to the power d, growing sharply as the distance increases.
C. 1 divided by d raised to the power q, so farther nodes are less likely.
D. d multiplied by q, the distance scaled by the clustering exponent.

Answer: C

Reference: Lectures 148–149 – Decentralized Search II–III (≈16:15–end)

Solution: The link probability is proportional to 1/d^q, so the chance of a long-range tie falls as the distance d grows — nearby far-nodes are more likely than very distant ones. The other forms make distant links more likely, grow with distance, or scale linearly.

Q4. [Arrange in order] Arrange the steps of one myopic (decentralized) search in the correct order.

Steps (in scrambled order):

(i) Among those neighbours, find the one whose lattice distance to the target is smallest.
(ii) Start at the source node holding the message.
(iii) Forward the message to that closest neighbour and make it the current node.
(iv) At the current node, look only at its own immediate neighbours.
(v) Repeat until the current node is the target, then report the path.

A. ii → iv → i → iii → v
B. iv → ii → i → iii → v
C. ii → i → iv → iii → v
D. ii → iv → iii → i → v

Answer: A (ii → iv → i → iii → v)

Reference: Lectures 147–148 – Decentralized Search I–II

Solution: Start at the source (ii), look only at the current node's own neighbours (iv), pick the neighbour closest to the target on the lattice (i), forward to it and advance (iii), and repeat until the target is reached (v).

Q5. [MCQ] What makes decentralized (myopic) search a “local” method?

A. Each node knows the entire global network and computes the true shortest path directly.
B. A central controller inspects the whole graph and tells each node exactly where to forward.
C. Each node knows only its own neighbours and forwards to the one closest to the target.
D. Each node forwards the message to a neighbour chosen uniformly at random each time.

Answer: C

Reference: Lecture 147 – Decentralized Search I (≈01:55–end)

Solution: Myopic search is local because each holder knows only its own contacts and greedily forwards to the neighbour it judges closest to the target — no global map and no central authority. It is neither globally informed, centrally directed, nor random.

Q6. [Statement type] Consider the two statements about myopic (decentralized) search and decide their truth.

Statement I: Myopic search always returns the globally shortest path between source and target.
Statement II: Myopic search uses only local information about the current node's own neighbours.

A. Both statements are true.
B. Both statements are false.
C. Statement I is true, but Statement II is false.
D. Statement I is false, but Statement II is true.

Answer: D

Reference: Lectures 147–148, 156 – Decentralized / myopic search

Solution: The first statement is wrong: being greedy and local, myopic search can return a longer path than the true shortest one. The second is correct: it relies only on the current node's own neighbours. So the first fails and the second holds.

Q7. [MCQ] When the current node estimates a neighbour's distance to the target, which links does it use?

A. Only the lattice (homophily) links, ignoring the neighbour's unknown long-range ties.
B. All links including the neighbour's long-range ties, which it always knows in full.
C. Only the neighbour's long-range ties, ignoring every lattice link in the network.
D. A randomly chosen half of all the links present in the entire network each time.

Answer: A

Reference: Lecture 155 – Myopic Search Introduction (≈03:13–end)

Solution: Distances are judged along the lattice only, because a node does not know its neighbours' long-range ties; that is why the search is short-sighted. It cannot use unknown long-range ties, cannot ignore the lattice, and does not sample links at random.

Q8. [MCQ] For efficient decentralized search on a two-dimensional grid, which clustering exponent works best?

A. An exponent of 0, making every far-away node equally likely to be linked.
B. An exponent of 4, so that almost no long-range links are ever formed.
C. An exponent of 1, which is the best value for a one-dimensional ring, not a grid.
D. An exponent of 2, balancing short-range and long-range links on the grid.

Answer: D

Reference: Lecture 149 – Decentralized Search III (≈20:43–end)

Solution: On a two-dimensional grid the search-path length is minimised when the clustering exponent equals 2, the value that balances links across all distance scales. An exponent of 0 makes far links uniform, 1 is the optimum for a one-dimensional ring, and 4 suppresses long-range links almost entirely.

Q9. [MCQ] As the network size n grows, how does the number of hops used by myopic search behave?

A. It grows in direct proportion to n, doubling whenever the network doubles in size.
B. It grows roughly like log n, increasing only slowly as the network gets larger.
C. It stays fixed at exactly six hops no matter how large the network becomes.
D. It grows like n squared, exploding far faster than the network itself grows.

Answer: B

Reference: Lecture 158 – Time Taken by Myopic Search (≈11:33–end)

Solution: Experiments show the myopic search path length grows roughly logarithmically with n — slowly — not linearly. It is neither a fixed six hops nor a quadratic blow-up.

Q10. [MSQ] Which statements about the model and its search are correct? (Select all that apply.)

I. Watts–Strogatz starts from a ring lattice and rewires a few edges.
II. Myopic search forwards to the neighbour closest to the target on the lattice.
III. On a 2-D grid the best clustering exponent for search is 2.
IV. Myopic search always finds the globally shortest path.

A. I, II, IV
B. II, III, IV
C. I, III
D. I, II, III

Answer: D (statements I, II, III)

Reference: Lectures 146–149, 156–158 – WS model and myopic search

Solution: The ring-lattice-plus-rewiring construction (I), greedy local forwarding (II), and the optimal grid exponent of 2 (III) are all correct. Myopic search is not guaranteed to be optimal, so IV is false.
